



Original article

***F2RL3* methylation in blood DNA is a strong predictor of mortality**

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Abstract

Background: Smoking is a major cause of morbidity and mortality. Smoking-related epigenetic biomarkers may open new avenues to better quantify the adverse health effects of smoking, and to better understanding of the underlying mechanisms. We aimed to evaluate the clinical implications of *F2RL3* methylation, a novel epigenetic biomarker of smoking exposure disclosed by recent genome-wide methylation studies.

Methods: Blood DNA methylation at *F2RL3* (also known as PAR-4) was quantified in baseline samples of 3588 participants aged 50–75 years in a large population-based prospective cohort study by MALDI-TOF mass spectrometry. Deaths were recorded during a median follow-up of 10.1 years. The associations of methylation intensity and of smoking with all-cause, cardiovascular, cancer and other mortality were assessed by Cox's proportional hazards regression, controlling for potential confounding factors.

Results: Lower methylation intensity at *F2RL3* was strongly associated with mortality. After adjustment for multiple covariates including smoking, hazard ratios [95% confidence interval (CI)] for death from any cause, cardiovascular disease, cancer or other causes were 2.60 (95% CI, 1.81–3.74), 2.45 (95% CI, 1.28–4.68), 2.94 (95% CI, 1.68–5.14) and 2.39 (95% CI, 1.11–5.16), respectively, in subjects in the lowest quartile of methylation intensity compared with subjects in the highest quartile. The associations with mortality outcomes were much stronger among men than among women. In addition, strong positive associations of smoking with each of the outcomes were substantially weakened, and almost disappeared when controlling for *F2RL3* methylation intensity.

Conclusions: *F2RL3* methylation is a strong predictor of mortality, including all-cause, cardiovascular, cancer and other mortality. Systemic adverse effects of smoking may be mediated by pathways associated with *F2RL3* methylation.

Key words: *F2RL3* methylation, PAR-4, smoking, mortality, prospective study

Key Messages

- The clinical implications of *F2RL3* methylation remained uncertain, and only one study reported that *F2RL3* methylation was associated with prognosis of patients with stable coronary heart disease.
- We found that *F2RL3* methylation strongly predicts the major smoking-associated outcomes, including all-cause mortality, as well as mortality from cardiovascular disease and cancer.
- *F2RL3* methylation might be involved in pathways mediating the systemic adverse effects of smoking.

Introduction

Tobacco smoking is and will continue to be the leading preventable cause of premature death worldwide.¹ It accounts for a large share of a variety of diseases,² including lung cancer (71%), chronic respiratory disease (42%) and cardiovascular disease (10%).³ According to a recent meta-analysis, current and former smokers have more than 80% and 30% increased all-cause mortality, respectively, compared with nonsmokers even in old age.⁴ How smoking imposes its adverse health effects on multiple human organs has been extensively studied,² but remains incompletely understood. Recent developments in research on epigenetic modification, such as DNA methylation, may open new avenues to better understanding of the underlying mechanisms, and better quantification of smoking-associated adverse health effects.

Recent genome-wide methylation studies have disclosed a strong inverse association between smoking and *F2RL3* (the coagulation factor II receptor-like 3 gene) methylation in blood DNA.^{5–8} *F2RL3* encodes for the protease-activated receptor-4 (PAR-4),⁹ which has been suggested to be involved in the pathophysiology of both the cardiovascular system^{10–15} and neoplastic diseases.^{16–22} These patterns suggest that the adverse health effects of smoking might be mediated in part by pathways going along with *F2RL3* methylation. To further explore this suggestion we assessed the association of *F2RL3* methylation intensity and of smoking with all-cause, cardiovascular, cancer and other mortality in a large cohort of older adults.

Materials and Methods

Study design and participants

The study subjects were drawn from the baseline population of the ESTHER study, a statewide population-based cohort study conducted in Saarland, Germany.²³ Briefly, 9949 participants aged 50–75 years (mean age 62 years) were recruited by their general practitioners during a general health check-up between July 2000 and December 2002, and followed up since then. The study was approved by the ethics committees of the medical faculty of the

University of Heidelberg, and the medical board of the State of Saarland, Germany. Written informed consent was obtained from each participant. Methylation at *F2RL3* in blood DNA was measured among participants recruited during the initial 9 months of the enrolment (between July 2000 and March 2001, $n=3624$), who were included in the current analysis.

Data collection

At baseline, a standardized self-administered questionnaire was completed by each participant, collecting information on socio-demographic characteristics, lifestyle factors, medical history, health status and history of major diseases. Prevalent diseases, such as diabetes or hypertension, were identified by documented diagnoses or prescriptions of pertinent medications in the medical records of the general practitioners. Prevalent cardiovascular disease at baseline was defined by either physician-reported coronary heart disease or a self-reported history of myocardial infarction, stroke, pulmonary embolism or revascularization of coronary arteries. Prevalent cancer at baseline was defined by a self-reported history of cancer. Furthermore, blood samples were taken during the health check-up, centrifuged and stored at -80°C until further processing. Total cholesterol level was measured by standard high-performance liquid chromatography methods. Deaths during follow-up between 2000 and the end of 2010 were identified by record linkage with population registries. Information on the major cause of death was obtained from death certificates provided by the regional public health offices, and coded with ICD-10 codes. Cardiovascular and cancer deaths were defined by ICD-10 codes I00–I99 and C00–C99, respectively.

Methylation assessment

DNA was extracted from whole blood samples collected at baseline using a salting out procedure.²⁴ Sequenom matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectrometry was used to quantify DNA methylation at an intragenic region within *F2RL3*.⁵ Briefly, genomic DNA samples (200 ng) were first

bisulfite-converted using the EZ-96 DNA Methylation Gold Kit (Zymo Research). Subsequently, the target region⁵ was amplified using bisulfite-specific primers [5'-aggaagagag GGTATTAGTAGTATGGTGGAGGG-3' (sense) and 5'-cagtaatacactactataggagaaggctACTTCTAACTAAATA CCCACCAA-3', uppercase letters indicate the sequence specific regions, and the nonspecific tags are shown in lower-case letters], followed by shrimp alkaline phosphatase (SAP) treatment, and RNase A cleavage (known as T-cleavage) performed according to the standard protocol (Sequenom EpiTyper Assay). The PCR product fragments were then cleaned by resin, and spotted on 384 SpectroCHIPS by Nanodispenser. The chip was analyzed by a Sequenom Autoflex Mass Spectrometer system, and data were extracted using SpectroACQUIRE v3.3.1.3 software and MassARRAY EpiTyper v1.0 software. The procedures described above are able to quantify the proportion of methylated DNA at four CpG sites (henceforth referred to as the methylation intensity at CpG_2 to CpG_5. CpG_1 could not be measured due to too low a mass in the MassArray; CpG_2 equals cg03636183 (corresponding numbers on the 27K and 450K assay), the locus identified as differentially methylated according to smoking exposure by genome-wide studies⁵⁻⁸). As SNPs at the primers' regions or at/near CpGs can influence methylation intensity, primers were designed excluding SNPs. Searching online databases also did not identify the presence of any SNPs within the target region. This assay showed high test-retest reliability and very limited well/position effects (measurements for 96 duplicate samples with Pearson correlation coefficients of 0.89-0.91, and mean difference $\leq 0.01\%$ 5mc for measurable CpGs). All the assays were performed by the same operator in the same laboratory. As procedures after bisulfite treatment were processed by batches corresponding to chips ($n = 11$), a variable representing the chip was defined as a batch indicator to be controlled for in the statistical analyses.

The methylation intensities at *F2RL3* CpG_2, CpG_4 and CpG_5 were highly correlated (all mutual Spearman correlation coefficients ≥ 0.84). CpG_3 cannot be well characterized by the MALDI-TOF assay due to low mass of the cleavage product. Its methylation could not be determined in 3.1% of samples, showed low test-retest reliability (Pearson correlation coefficients = 0.56), and lower correlation with methylation at the other CpG sites (Spearman correlation coefficients, 0.32-0.33), as previously described.^{5,25} This site was therefore not further considered in the analyses.

Statistical analysis

The study population was first characterized with respect to major sociodemographic characteristics, lifestyle factors

and prevalent diseases. Median and interquartile methylation at target CpGs of *F2RL3* was determined according to levels of these characteristics, and differences in methylation were examined by Kruskal-Wallis test.

Kaplan-Meier plots and log-rank tests were employed to assess associations of methylation intensity categorized by quartiles with all-cause and cause-specific (cardiovascular, cancer, other) mortality. In addition, overall and sex-specific hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated by Cox regression models, first adjusting for age, sex and batch effect only (Model I); further adjustments additionally included smoking status (never/former/current smoker) (Model II); and finally the following established risk factors (Model III / fully adjusted model): body mass index (kg/m^2), alcohol consumption (g/day), physical activity (inactive, low, medium/high), systolic blood pressure (mmHg), total cholesterol level (mg/dl), prevalent hypertension, cardiovascular disease, diabetes and cancer. Methylation intensities were entered either as quartiles (highest quartile defined as the reference category) or as continuous variables (calculating hazard ratios for 10% units lower proportion of methylated DNA). In addition, restricted cubic spline regression was employed to explore the potential non-linearity of the association between methylation intensity and survival experience.²⁶ To evaluate potential variation of the association between *F2RL3* methylation and mortality with respect to time, all analyses were repeated with follow-up time restricted to 5 and 8 years, respectively.

The proportional hazards assumption was assessed by martingale-based residuals.^{27,28} Potential interactions between methylation and the covariates were tested for statistical significance by adding pertinent product terms (methylation quartiles*sex, methylation quartiles*smoking status, methylation quartiles*age) in the fully adjusted models.

In sensitivity analyses, models for cardiovascular mortality were restricted to subjects without cardiovascular disease at baseline, models for cancer mortality were repeated after excluding subjects with a life-time history of cancer and those who died of cancer in the first five years of follow-up, and models for all-cause mortality were repeated after excluding both subsets of aforementioned subjects. The rationale behind those analyses was to explore potential bias due to reverse causality. A further sensitivity analysis was carried out in which C-reactive protein (CRP), an inflammatory marker that has been reported to be associated with both absolute white blood cell (WBC) count and WBC type distribution,²⁹ was included in the full model.

In additional analyses, the associations between smoking at baseline and all mortality endpoints were estimated

by Cox regression as well, with and without control for methylation intensities. These analyses were carried out to compare the associations between *F2RL3* methylation and mortality with the associations between smoking and mortality, in order to explore to what extent the associations of smoking with mortality might be mediated by pathways associated with *F2RL3* methylation. Further analyses were performed by classifying the study population according to both smoking status and methylation intensity, and using the never smokers in the highest quartile of methylation level as reference group.

All aforementioned analyses were first done separately for CpG_2, CpG_4, and CpG_5, and then repeated using aggregate measures of methylation intensity (the average of the CpG sites). However, results of the latter were highly consistent with those for the individual CpGs, and are therefore not reported separately. Results for CpG_4 are presented in the main text, and results for CpG_2 and CpG_5 are provided in the [Supplementary Appendix](#), available as [Supplementary data](#) at *IJE* online.

All data analyses were conducted using SAS 9.2 (SAS Institute, Cary, NC), and all statistical tests were two-sided with *P*-values of <0.05.

Results

Of 3624 participants recruited in the ESTHER study between July 2000 and March 2001, mortality follow-up was available for 3623 subjects (>99.9%), and methylation levels could be determined at one or more CpG sites for 3589 (99.0 %) subjects, which allowed inclusion of 3588 subjects in the analysis. Characteristics of the study population at baseline are shown in [Table 1](#). Mean age was 62 years; the sample included slightly more women than men. More than half of participants had ever smoked, close to 20% were still smoking at the time of recruitment. During a median follow-up time of 10.1 years, 469 participants died. Among 452 participants for whom death certificates could be obtained (96.4%), 151 died from cardiovascular disease, 197 died from cancer and 105 died from other causes.

[Table 1](#) also shows methylation intensities at *F2RL3* CpG_4 by sex, age and smoking status (corresponding results for the other CpGs are reported in [Supplementary Table S1](#), available as [Supplementary data](#) at *IJE* online). Median methylation was somewhat lower among men than among women but rather consistent across age groups. By far the strongest variation of methylation intensity was seen according to smoking status. Compared with participants who never smoked, current smokers had much lower methylation levels. Intermediate levels (closer to those of never smokers) were observed for former smokers.

Table 1. Baseline characteristics and *F2RL3* (CpG_4) methylation intensity of the study population

Characteristics	No. (%)	Methylation intensity		
		Median	(Q1-Q3)	P-value ^a
Overall	3588 (100)	0.79	(0.72 – 0.84)	
Sex				
Male	1594 (44.4)	0.77	(0.66 – 0.82)	
Female	1994 (55.6)	0.80	(0.75 – 0.84)	<0.0001
Age (years)				
<60	1265 (35.3)	0.79	(0.69 – 0.84)	
60–64	1025 (28.6)	0.80	(0.72 – 0.84)	
65–69	789 (22.0)	0.79	(0.73 – 0.84)	
70–75	509 (14.2)	0.79	(0.72 – 0.84)	0.04
Smoking status ^b				
Never smoker	1701 (48.7)	0.82	(0.78 – 0.85)	
Former smoker	1136 (32.5)	0.77	(0.70 – 0.82)	
Current smoker	654 (18.7)	0.62	(0.53 – 0.73)	<0.0001

Q1, 1st quartile; Q3, 3rd quartile.

^aKruskal–Wallis test for group differences.

^bData missing for 97 subjects.

Moreover, methylation intensity showed strong inverse associations with smoking intensity among current smokers and with pack-years of smoking among ever smokers, and strong positive associations with years since cessation among ever smokers (all *P*-values <0.0001. Detailed relationships between smoking behaviour and *F2RL3* methylation have been reported elsewhere³⁰).

[Figure 1](#) depicts the survival experience according to quartiles of *F2RL3* CpG_4 methylation with respect to deaths from all causes, cardiovascular disease, cancer and other causes (corresponding results for the other CpGs are illustrate in [Figure S1](#), available as [Supplementary data](#) at *IJE* online). A strong gradient of lower survival among participants with lower methylation levels was consistently observed.

The associations of methylation intensities at *F2RL3* CpG_4 with all-cause and cause-specific mortality are further presented in [Table 2](#). Risk of death from any cause monotonously increased with decreasing methylation intensity, showing a 2.9-fold hazard for the lowest methylation quartile in comparison with the highest quartile. Inclusion of smoking status only slightly attenuated the hazard ratio (HR) estimate after adjustment for age and sex. Additional control for various other potential confounding factors likewise only slightly reduced the observed association (HR = 2.60; 95% CI, 1.81–3.74). Similar hazard ratios were observed for cardiovascular, cancer and other mortality. In models including methylation intensity as a linear term, a decrease in methylation by 10% units was associated with an increase in all-cause

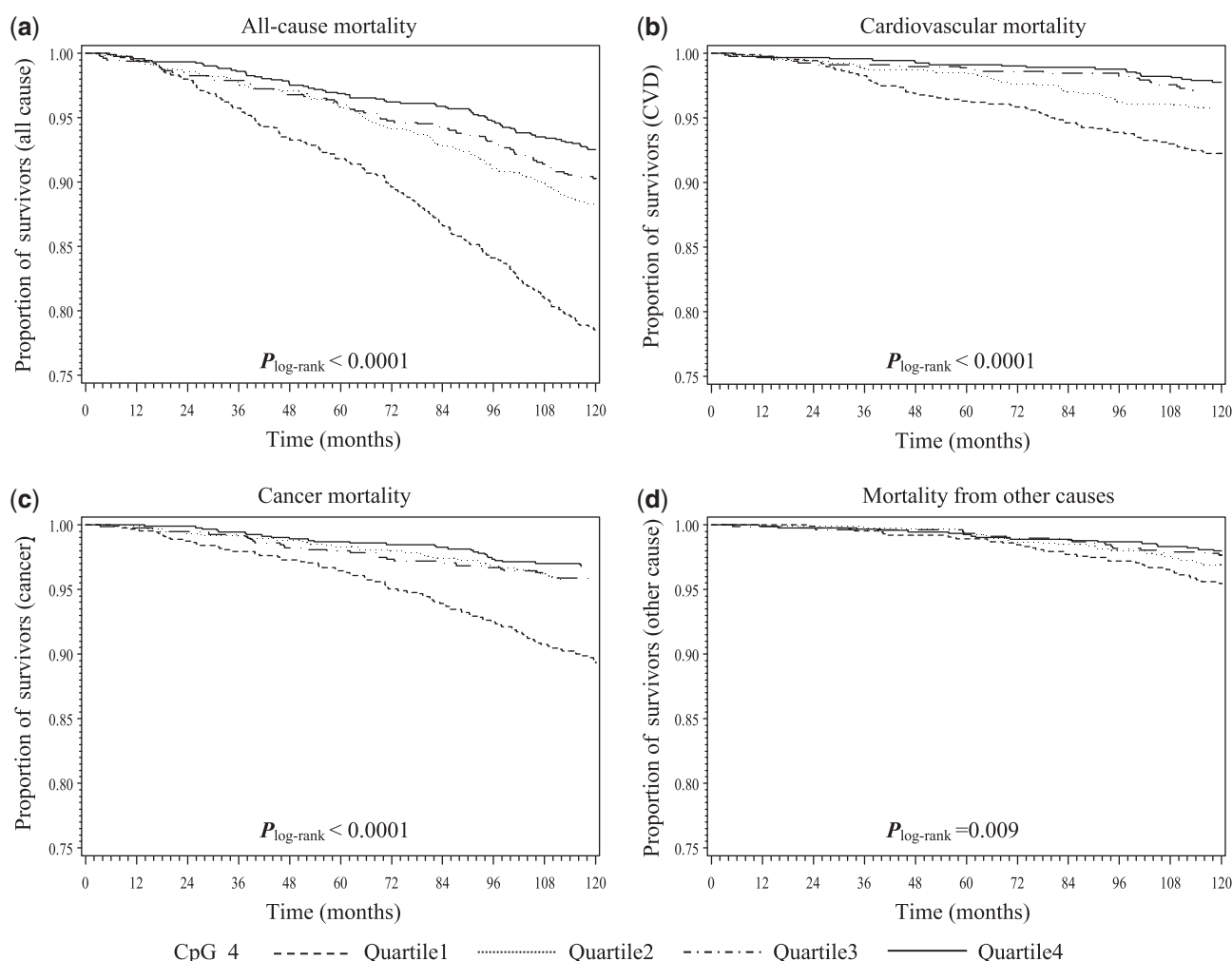


Figure 1. Kaplan-Meier estimates of survival with respect to death from all causes (a), cardiovascular disease (b), cancer (c) and other causes (d) by *F2RL3* CpG_4 methylation quartiles.

mortality and mortality from cardiovascular disease, cancer and other causes by 37%, 38%, 43%, and 33%, respectively, in fully adjusted models. Corresponding results for other CpGs are reported in the [Supplementary Table S2](#), available as [Supplementary data](#) at *IJE* online. Furthermore, dose-response analyses by restricted cubic spline regression indicated a monotonic inverse association between *F2RL3* methylation and the various outcomes (Figure 2, and Figure S2 available as [Supplementary data](#) at *IJE* online). The strength of the observed associations was quite consistent for various lengths of follow-up (Figure 3, and Figure S3 available as [Supplementary data](#) at *IJE* online).

Standardized score process plots indicated no violation of the proportional hazard assumption for any of the Cox regression models. Statistical interactions were detected between methylation quartiles and sex for cancer mortality ($P=0.02$), and all-cause mortality ($P=0.03$). Sex-specific analyses showed the strength of the associations to

be more pronounced among men than among women (Table 3, and Table S3 available as [Supplementary data](#) at *IJE* online).

In sensitivity analyses excluding participants with a history of cardiovascular disease or cancer, and participants who died of cancer in the first 5 years of follow-up, the observed associations became even stronger. For example, the HRs for the lowest vs highest quartile of *F2RL3* CpG_4 methylation increased to 3.17 (95% CI, 1.84-5.44), 2.94 (95% CI, 1.35-6.43) and 4.91 (95% CI, 2.22-10.86) for all-cause, cardiovascular and cancer mortality, respectively, in the fully adjusted models. Adjusting for CRP in the full model in additional sensitivity analyses did not materially alter any of the associations (data not shown).

Current smoking was also strongly associated with all mortality endpoints in age-, and sex-adjusted analyses. However, the associations were strongly attenuated or even almost disappeared (cardiovascular mortality) after

Table 2. Association of methylation intensity at *F2RL3* (CpG4) and smoking with all-cause and cause specific mortality

Outcome	Methylation level/ smoking status	<i>N</i> _{total} ^a	Cases	PY	IR ^b	HR (95% CI)		
						Model 1 ^c	Model 2 ^d	Model 3 ^e
All-cause mortality	0.84 – 1.00 (Quartile 4)	927	71	9108.05	0.78	Ref.	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	805	83	7761.20	1.07	1.30 (0.95 – 1.78)	1.37 (0.99 – 1.89)	1.35 (0.94 – 1.93)
	0.72 – 0.79 (Quartile 2)	961	114	9203.15	1.24	1.43 (1.06 – 1.92)	1.45 (1.07 – 1.98)	1.32 (0.93 – 1.87)
	0.33 – 0.71 (Quartile 1)	895	201	8197.92	2.45	2.91 (2.21 – 3.84)	2.75 (1.99 – 3.79)	2.60 (1.81 – 3.74)
	Per 10% less methylation					1.40 (1.30 – 1.50)	1.36 (1.24 – 1.48)	1.37 (1.23 – 1.51)
	Never smoker	1701	168	16496.35	1.02	Ref.	Ref.	Ref.
	Former smoker	1136	165	10735.39	1.53	1.26 (1.00 – 1.59)	1.00 (0.78 – 1.28)	0.93 (0.71 – 1.22)
	Current smoker	654	117	6150.36	1.90	2.11 (1.65 – 2.70)	1.21 (0.89 – 1.63)	1.30 (0.93 – 1.80)
Cardiovascular mortality	0.84 – 1.00 (Quartile 4)	925	21	9100.50	0.23	Ref.	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	800	23	7730.15	0.30	1.20 (0.66 – 2.17)	1.21 (0.67 – 2.19)	1.08 (0.55 – 2.11)
	0.72 – 0.79 (Quartile 2)	956	40	9174.60	0.44	1.64 (0.96 – 2.78)	1.56 (0.91 – 2.67)	1.37 (0.74 – 2.53)
	0.33 – 0.71 (Quartile 1)	891	67	8174.50	0.82	3.21 (1.94 – 5.30)	3.01 (1.71 – 5.29)	2.45 (1.28 – 4.68)
	Per 10% less methylation					1.42 (1.26 – 1.60)	1.41 (1.21 – 1.66)	1.38 (1.14 – 1.66)
	Never smoker	1701	55	16460.62	0.33	Ref.	Ref.	Ref.
	Former smoker	1136	57	10716.30	0.53	1.32 (0.88 – 1.97)	1.00 (0.65 – 1.53)	0.92 (0.56 – 1.49)
	Current smoker	654	32	6114.61	0.52	1.83 (1.16 – 2.89)	0.96 (0.56 – 1.66)	1.08 (0.58 – 1.98)
Cancer mortality	0.84 – 1.00 (Quartile 4)	925	29	9100.50	0.32	Ref.	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	800	36	7730.15	0.47	1.39 (0.85 – 2.27)	1.61 (0.97 – 2.69)	1.55 (0.90 – 2.67)
	0.72 – 0.79 (Quartile 2)	956	40	9174.60	0.44	1.25 (0.77 – 2.01)	1.37 (0.82 – 2.27)	1.17 (0.68 – 2.03)
	0.33 – 0.71 (Quartile 1)	891	92	8174.50	1.13	3.25 (2.12 – 4.98)	3.28 (1.98 – 5.44)	2.94 (1.68 – 5.14)
	Per 10% less methylation					1.45 (1.31 – 1.61)	1.42 (1.24 – 1.63)	1.43 (1.22 – 1.67)
	Never smoker	1701	65	16460.62	0.39	Ref.	Ref.	Ref.
	Former smoker	1136	71	10716.30	0.66	1.39 (0.96 – 1.99)	1.06 (0.72 – 1.56)	0.94 (0.62 – 1.41)
	Current smoker	654	54	6114.61	0.88	2.42 (1.66 – 3.54)	1.25 (0.79 – 1.99)	1.29 (0.79 – 2.09)
Other mortality	0.84 – 1.00 (Quartile 4)	925	19	9100.50	0.21	Ref.	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	800	19	7730.15	0.25	1.13 (0.60 – 2.13)	1.06 (0.55 – 2.07)	1.06 (0.48 – 2.33)
	0.72 – 0.79 (Quartile 2)	956	29	9174.60	0.32	1.38 (0.77 – 2.47)	1.38 (0.76 – 2.52)	1.49 (0.74 – 3.01)
	0.33 – 0.71 (Quartile 1)	891	38	8174.50	0.46	2.13 (1.21 – 3.75)	1.94 (1.00 – 3.77)	2.39 (1.11 – 5.16)
	Per 10% less methylation					1.28 (1.09 – 1.49)	1.22 (1.00 – 1.49)	1.33 (1.05 – 1.67)
	Never smoker	1701	48	16460.62	0.25	Ref.	Ref.	Ref.
	Former smoker	1136	37	10716.30	0.32	1.09 (0.66 – 1.78)	0.93 (0.55 – 1.55)	0.93 (0.52 – 1.67)
	Current smoker	654	31	6114.61	0.41	1.92 (1.14 – 3.23)	1.31 (0.69 – 2.47)	1.40 (0.69 – 2.85)

CI, confidence interval; HR, hazard ratio; PY, person-years; IR, incidence rate; Ref., reference category.

Bold printed: $P < 0.05$.^aSum does not add up to 3588 for the following reasons: 1 subject with unavailable methylation intensity at CpG_4, 16 subjects with unclear cause of death, and 97 subjects with missing smoking status; The unequal frequencies among quartiles occurred because methylation intensity was available with two decimals only.^bIncidence rate per 100 person-years.^cModel 1: adjusted for age, sex, and batch effect.^dModel 2: like model 1, additionally adjusted for smoking status/methylation intensity (quartiles).^eModel 3: like model 2, additionally adjusted for body mass index, alcohol consumption, physical activity, systolic blood pressure, total cholesterol, hypertension, cardiovascular disease, diabetes and cancer.

adjustment for *F2RL3* methylation (Table 2, and Table S2 available as [Supplementary data](#) at *IJE* online). Potential interactions between smoking and methylation level were tested by adding product terms of smoking categories and

methylation quartiles, yielding P -values >0.05 for all outcomes. In the analyses with joint classification of participants by both smoking status and methylation intensity, mortality was consistently increased in those with lowest

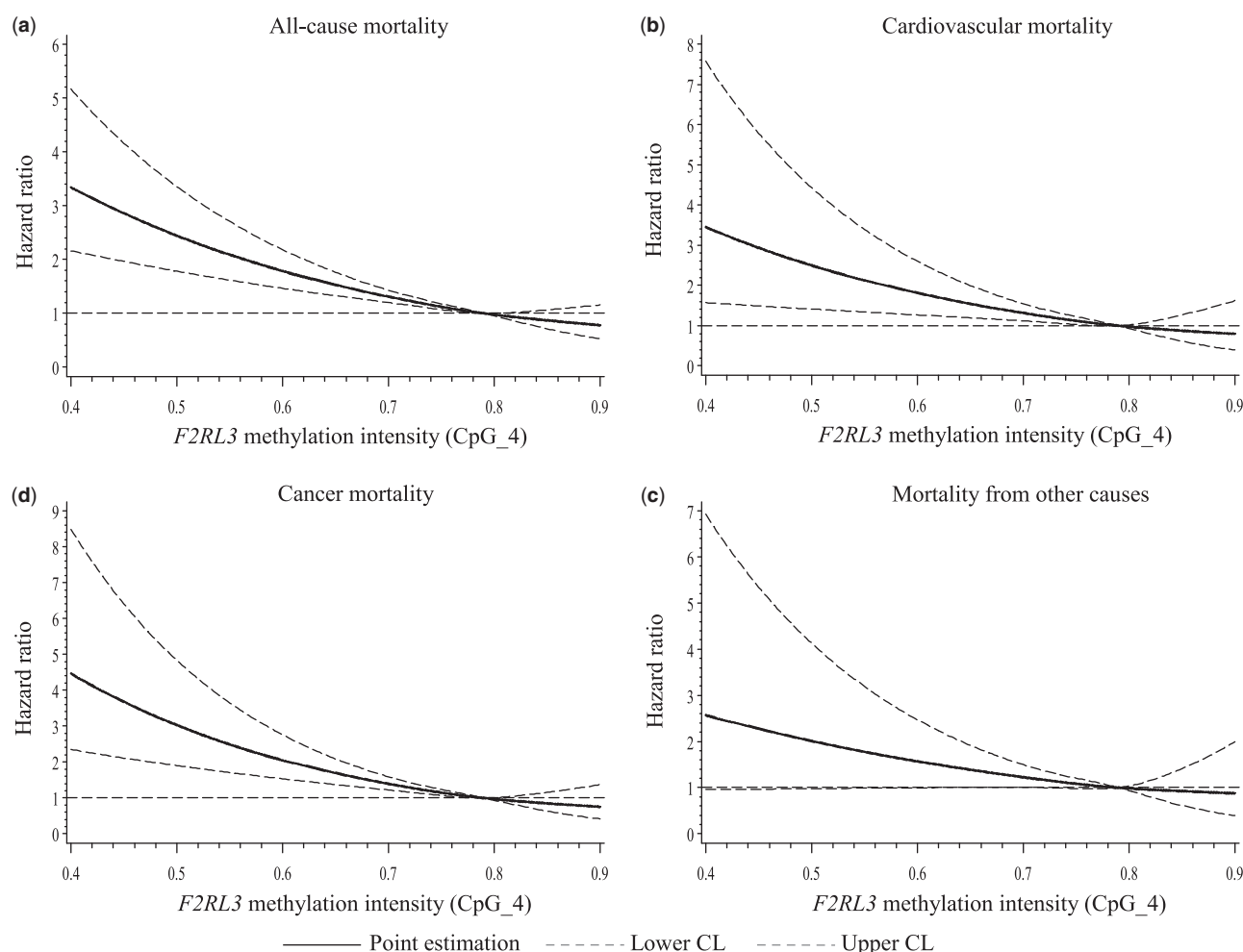


Figure 2. Dose-response relationship of *F2RL3* (CpG_4) methylation intensity with all-cause mortality (a), cardiovascular mortality (b), cancer mortality (c) and mortality of other causes (d) [results from restricted cubic spline multiple regression (HR estimated using the median 0.79 as reference), adjusted for potential confounding factors].

methylation levels in never smokers, former smokers and current smokers. (Table S4, available as [Supplementary data](#) at *IJE* online).

Discussion

In this large population-based cohort study with a median follow-up of 10.1 years, hypomethylation at *F2RL3* was associated with strongly increased all-cause, cardiovascular, cancer and other mortality, even after controlling for a broad range of risk factors, including smoking exposure. Furthermore, *F2RL3* methylation appeared to account for a substantial proportion of smoking-related excess mortality. Our results are in agreement with and strongly expand previous findings of a strong inverse association of *F2RL3* methylation with all-cause, cardiovascular and non-cardiovascular mortality recently reported in a cohort of 1206 patients with stable coronary heart disease.²⁵ The current results, which are based on a much larger cohort from the

general population, indicate that this strong and consistent association is not restricted to this specific patient group.

The *F2RL3* gene which codes for the thrombin protease-activated receptor-4 (PAR-4) has been found to play a role in mediating platelet activation^{10–12} and a multitude of signalling pathways relevant to endothelial cell functions, and inflammatory reactions in the vascular system.^{12–15} Of note, these pathological features have likewise been described among smoking-induced detrimental effects.^{31,32} PAR-4 was also reported to be involved in the development of various premalignant and malignant conditions, including lung fibrosis,¹⁶ ulcerative colitis,¹⁷ colon cancer,¹⁸ prostate cancer,¹⁹ and gastric cancer²⁰ as well as metastasis.^{21,22} Recently, a very strong association of *F2RL3* methylation intensity in blood DNA with smoking exposure, a well-established risk factor of cardiovascular disease and cancer as well as all-cause mortality, has been discovered by genome-wide methylation studies.^{5–8} It has also been suggested that *F2RL3* expression tentatively

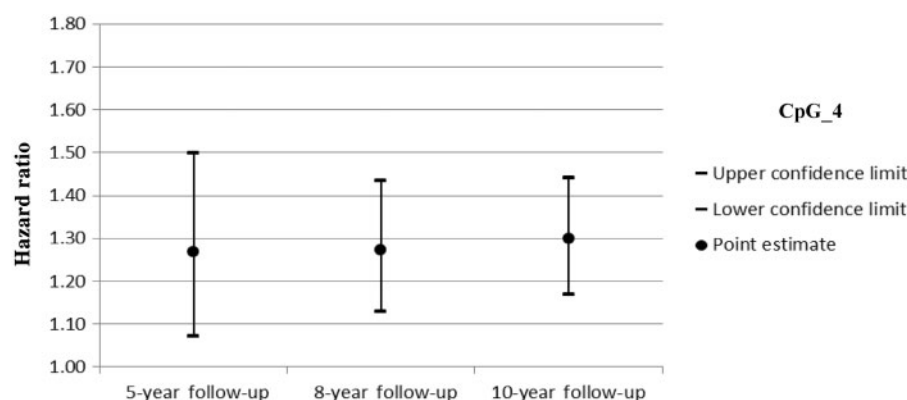


Figure 3. Associations between 10% less methylation intensity at *F2RL3* (CpG_4) and all-cause mortality by follow-up time

Table 3. Sex-specific association between methylation intensity at *F2RL3* (CpG4) and mortality outcomes

Outcome	Methylation level ^a	HR (95% CI) ^b	
		Females	Males
All-cause mortality	0.84 – 1.00 (Quartile 4)	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	1.14 (0.72 – 1.82)	1.94 (1.06 – 3.55)
	0.72 – 0.79 (Quartile 2)	0.71 (0.43 – 1.18)	2.66 (1.51 – 4.70)
	0.33 – 0.71 (Quartile 1)	2.13 (1.21 – 3.76)	4.08 (2.31 – 7.21)
	Per 10% less methylation	1.26 (1.04 – 1.53)	1.42 (1.26 – 1.61)
Cardiovascular mortality	0.84 – 1.00 (Quartile 4)	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	0.97 (0.37 – 2.50)	1.28 (0.47 – 3.50)
	0.72 – 0.79 (Quartile 2)	0.64 (0.24 – 1.73)	2.21 (0.91 – 5.39)
	0.33 – 0.71 (Quartile 1)	1.98 (0.66 – 5.91)	3.17 (1.28 – 7.81)
	Per 10% less methylation	1.20 (0.85 – 1.69)	1.48 (1.19 – 1.85)
Cancer mortality	0.84 – 1.00 (Quartile 4)	Ref.	Ref.
	0.80 – 0.83 (Quartile 3)	1.67 (0.86 – 3.22)	2.19 (0.77 – 6.24)
	0.72 – 0.79 (Quartile 2)	0.61 (0.27 – 1.37)	3.12 (1.17 – 8.33)
	0.33 – 0.71 (Quartile 1)	2.62 (1.13 – 6.09)	5.66 (2.12 – 15.06)
	Per 10% less methylation	1.38 (1.03 – 1.85)	1.45 (1.21 – 1.75)

CI, confidence interval; HR, hazard ratio.

Bold printed: $P < 0.05$.

^aQuartile cut-points were determined from the overall population.

^bAdjusted for age, smoking status, body mass index, alcohol consumption, physical activity, systolic blood pressure, total cholesterol, hypertension, cardiovascular disease, diabetes, cancer and batch effect.

increases with duration of smoking exposure in animal models, even though the result was not statistically significant ($P > 0.05$), given the small sample size (five mice) and the short observation period (28 days).⁷ In the light of these findings, the observed pronounced associations of *F2RL3* methylation with various fatal endpoints are not unexpected. One plausible explanation might be that smoking-induced hypomethylation at *F2RL3* may upregulate the expression of PAR-4 observed in cardiovascular and malignant pathology.^{11–19,21,22} Although causal

pathways remain to be clarified, thrombin-associated systemic or local alteration in blood coagulation would appear to be a plausible explanation for the associations of *F2RL3* methylation with cardiovascular and cancer mortality.^{33–35} In addition, both *F2RL3* methylation and methylation at other intragenic and intergenic regions, such as *AHRR*, *ALPPL2* and *6p21.33*, have been shown to be highly correlated with smoking exposure,^{7,36} and the observed prospective associations may also partly reflect the role of other smoking-related epigenetic changes.

For instance, methylation at *F2RL3*, *AHRR*, *ALPPL2* and *6p21.33* were all associated with 4-vinylphenol, a metabolic indicator of complex disorders.³⁷ Further research is warranted to address the individual and joint associations of smoking-related DNA methylation patterns with mortality and the underlying mechanisms.

To our knowledge, no previous single biomarker has shown comparably strong associations with all of the leading causes of mortality. The magnitude of the associations of *F2RL3* methylation with deaths from all causes, cardiovascular disease, cancer and other causes was strikingly much larger than what is commonly seen in molecular and genetic epidemiological studies of complex diseases. Given the strong association with smoking behaviour,^{5–8} *F2RL3* methylation could be an extraordinarily accurate predictor of smoking-associated risk.

A large number of longitudinal epidemiological studies have consistently shown strong associations of smoking with all-cause, cardiovascular and cancer mortality.^{4,38,39} Although such associations were also found in our study, they were substantially attenuated or even disappeared after adjustment for *F2RL3* methylation intensity. This pattern would be consistent with the hypothesis that the detrimental effects of smoking may predominantly be mediated by methylation-associated pathways, including *F2RL3*, and/or other more recently identified smoking-associated loci, such as *AHRR*, *ALPPL2* and *6p21.33*.^{7,36} An alternative or complementary explanation may be that *F2RL3* methylation may more accurately reflect the biologically effective current and lifetime dose of smoking than self-reported smoking history and habits that might suffer from wilful misreporting and imperfect recall. The strong reduction of the associations of smoking with the mortality outcomes should not be misinterpreted, however, as an indication that smoking is not harmful by itself and that its apparent association with increased mortality might be due to some *F2RL3* methylation-associated confounding factors.

In our study, we found *F2RL3* methylation to be lower among men than among women and the association between *F2RL3* methylation and mortality to be much stronger among men than among women. The sex differences in *F2RL3* methylation might be explained by the much higher proportions of ever smokers among men (71.8%) than among women (34.7%) and higher smoking intensity in male smokers than in female smokers [e.g. median and interquartile range of lifetime pack-years were 28.4 (16.0–30.2) and 20.0 (8.8–21.1) for ever smoking men and women, respectively; $P < 0.0001$]. A plausible explanation for the stronger association of *F2RL3* methylation with mortality might be the much larger share of smoking-attributable cardiovascular and cancer deaths among

men than among women.^{40,41} For example, lung or head and neck cancer, which are most strongly related to smoking,^{3,41} accounted for 36.4% of all cancer deaths among men, compared with 20.3% among women in our study. Likewise, 58.2 % of cardiovascular deaths among men were caused by ischaemic heart disease (IHD), a cardiovascular condition ranked first among smoking-attributable cardiovascular mortality,⁴⁰ compared with 38.3% of cardiovascular deaths due to IHD among women in the current study population.

Our study has specific strengths and limitations. Strengths include the longitudinal study design with comprehensive long-term mortality follow-up and detailed information on a wide range of covariates in a large population-based cohort. Limitations include the limited numbers of cause-specific deaths that hampered more detailed analyses by more specific causes of deaths, such as deaths from specific types of cancer and from respiratory disease. Methylation intensities were only measured in baseline blood samples. Possible changes in methylation intensity during follow-up, e.g. in response to changes in smoking habits, could therefore not be considered. *F2RL3* methylation intensities measured in our study essentially refer to all types of leukocytes combined. Although methylation intensity may strongly vary between cell types,^{42,43} no variation of *F2RL3* methylation between various types of leukocytes was found in a recent study among healthy subjects.⁷ Moreover smoking, the main determinant of *F2RL3* methylation intensities, does not seem to affect the composition of leukocytes to a relevant extent, even though it raises some specific type of leukocytes more than others.⁴⁴ Furthermore, additional adjustment for CRP, an inflammatory marker known to be related to both absolute WBC count and WBC type distribution,²⁹ did not materially alter any of the HR estimates. We additionally calculated the potential difference of methylation at cg03636183 (the locus within *F2RL3* measured by the 450K beadchip and corresponding to CpG2 in the current study) with respect to whole blood, mononuclear cells and granulocytes in data published by Reinius et al.⁴⁵ and no differences were identified (Kruskal–Wallis test, $P = 0.30$). Therefore, *F2RL3* methylation-related mortality differences observed in our study are unlikely to reflect differential distribution of various types of leukocytes. Furthermore, even if the associations were fully or partly explained by differences in leukocyte type distribution, this would not diminish the value of *F2RL3* methylation as a smoking-related strong predictor of all-cause and cause-specific mortality as shown in our analyses.

Despite its limitations, our study demonstrates that methylation of *F2RL3* in blood DNA is a strong predictor of all-cause, cardiovascular, cancer and other mortality.

Our analyses furthermore suggest that *F2RL3* methylation might be involved in pathways mediating the detrimental effects of smoking on mortality. Further multidisciplinary research is needed to elucidate the underlying mechanisms, including eventual possibilities of intervention to limit the detrimental effects of smoking.

Supplementary Data

Supplementary data are available at *IJE* online.

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